Compiler Construction Lab - Assignment 11

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Batch: B3

Objective

To write a program that generates intermediate code for a given arithmetic expression containing the operators +, -, *, and /.

Theory

Intermediate code generation is a phase of compiler design that converts a source program into an intermediate representation (IR) before generating machine code. It makes the compiler more modular and simplifies optimization and code generation.

For arithmetic expressions, one common intermediate representation is Three-Address Code (TAC). In TAC, each instruction typically contains at most one operator, with the result stored in a temporary variable.

For example, the expression:

a + b * c is

converted into:

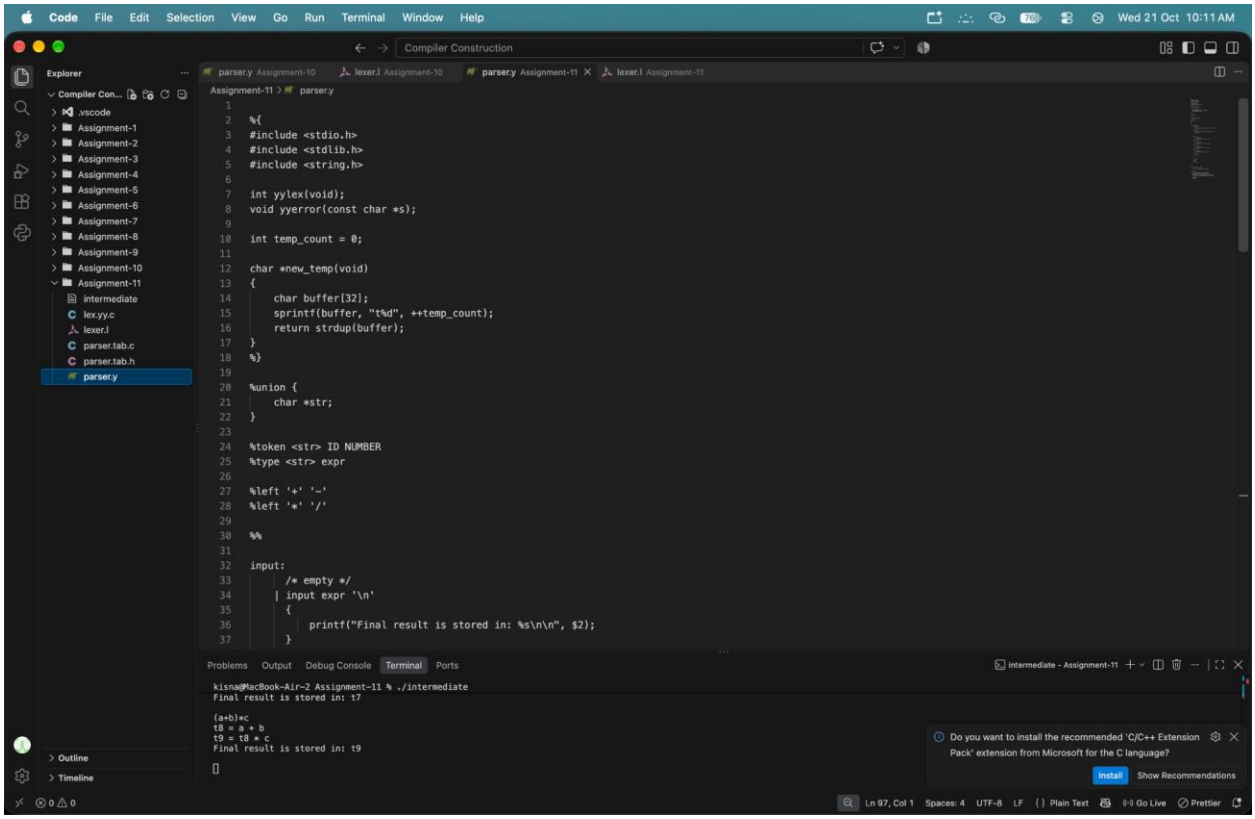
t1 = b * c t2

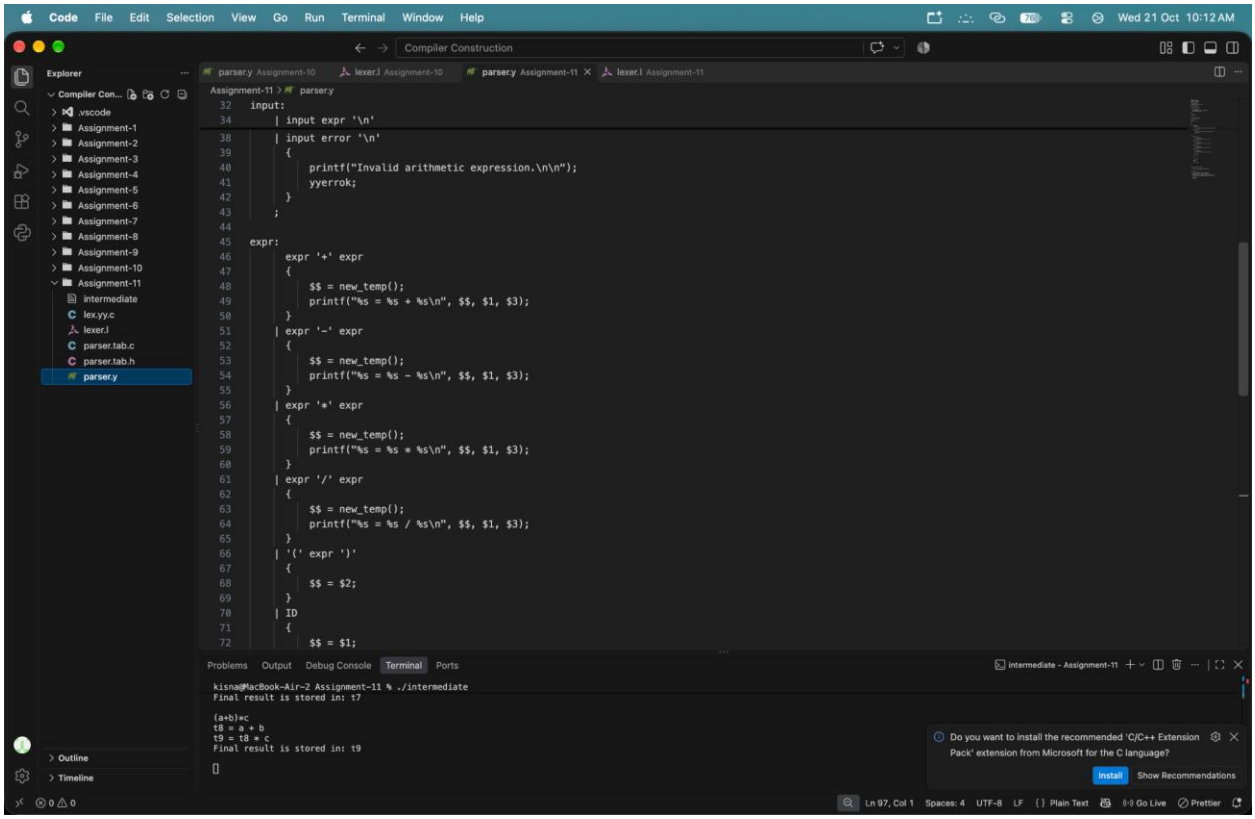
= a + t1

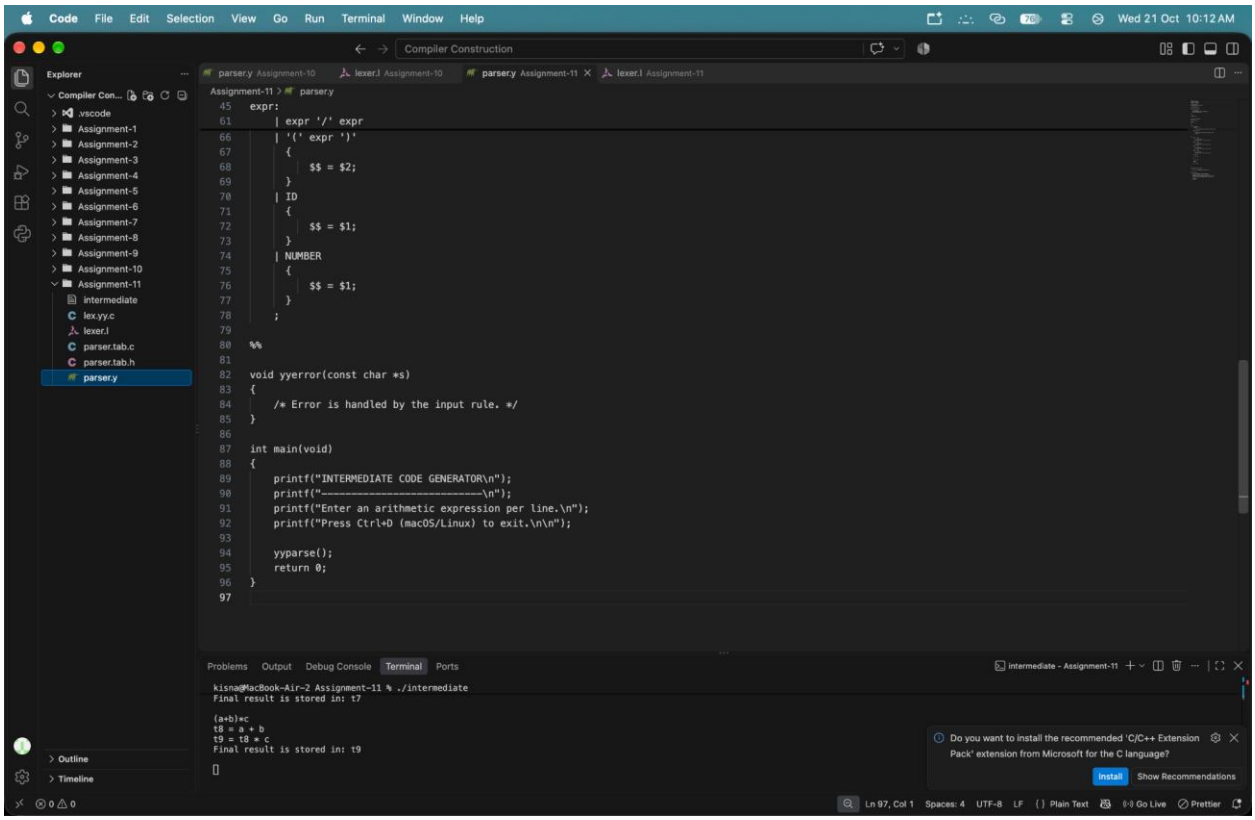
The multiplication is performed first because it has higher precedence than addition. Parentheses can be used to control the order of evaluation.

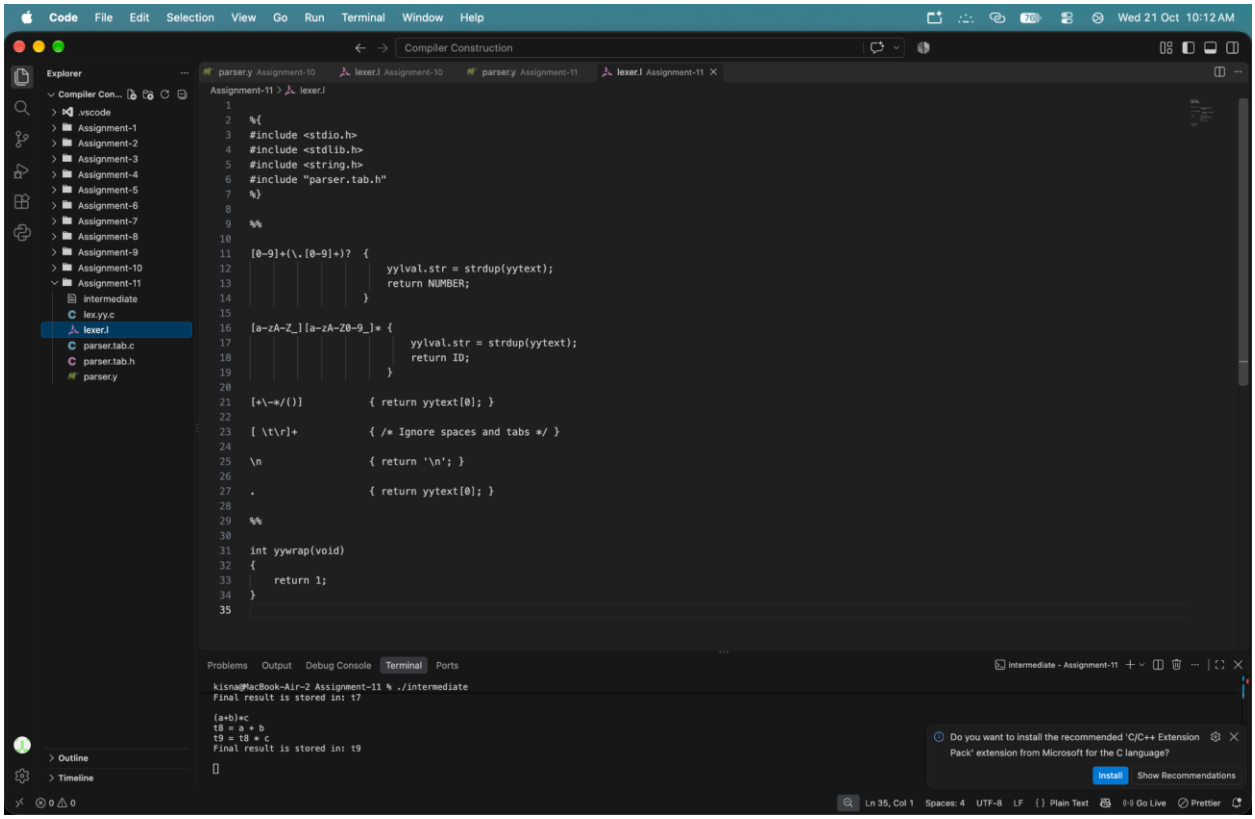
In this assignment, Flex recognizes identifiers, numbers, arithmetic operators, and parentheses, while Bison defines the grammar and generates three-address intermediate code using temporary variables such as t1, t2, and t3.

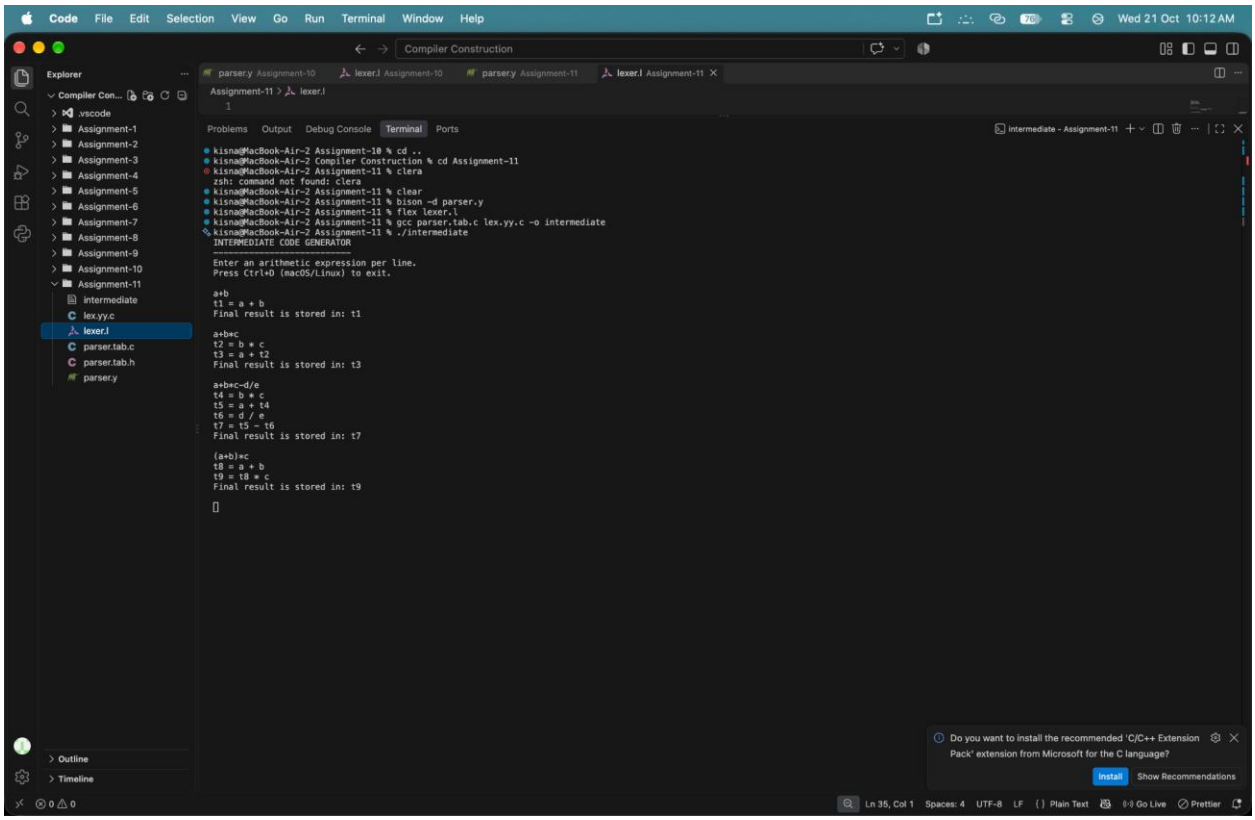
Screenshots and Execution Results




Code File Edit Selection View Go Run Terminal Window Help
Wed 21 Oct 10:12 AM
Compiler Construction
Explorer
parser.y Assignment-10
lexer.l Assignment-10
parsery Assignment-11
lexer.l Assignment-11
Assignment-11 > parser.y
Compiler Con...
.vscode
Assignment-1
Assignment-2
Assignment-3
Assignment-4
Assignment-5
Assignment-6
Assignment-7
Assignment-8
Assignment-9
Assignment-10
Assignment-11
intermediate
lex.yy.c
lexer.l
parser.tab.c
parser.tab.h
parser.y
input:
| input expr '\n'
| input error '\n'
{
printf("Invalid arithmetic expression.\n\n");
yyerrok;
}
;
expr:
expr '+' expr
{
$$ = new_temp();
printf("%s = %s + %s\n", $$, $1, $3);
}
| expr '-' expr
{
$$ = new_temp();
printf("%s = %s - %s\n", $$, $1, $3);
}
| expr '*' expr
{
$$ = new_temp();
printf("%s = %s * %s\n", $$, $1, $3);
}
| expr '/' expr
{
$$ = new_temp();
printf("%s = %s / %s\n", $$, $1, $3);
}
| '(' expr ')'
{
$$ = $2;
}
| ID
{
$$ = $1;
Problems Output Debug Console Terminal Ports
intermediate - Assignment-11
kisna@MacBook-Air-2 Assignment-11 % ./intermediate
Final result is stored in: t7
(a+b)*c
t8 = a + b
t9 = t8 * c
Final result is stored in: t9
Do you want to install the recommended 'C/C++ Extension Pack' extension from Microsoft for the C language?
Install
Show Recommendations
Outline
Timeline
Ln 97, Col 1
Spaces: 4
UTF-8
LF
Plain Text
Go Live
Prettier


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Assignment-8
Assignment-9
Assignment-10
Assignment-11
intermediate
lex.yy.c
lexer.l
parser.tab.c
parser.tab.h
parser.y
expr:
| expr '/' expr
| '(' expr ')'
{
$$ = $2;
}
| ID
{
$$ = $1;
}
| NUMBER
{
$$ = $1;
}
;
%%
void yyerror(const char *s)
{
/* Error is handled by the input rule. */
}
int main(void)
{
printf("INTERMEDIATE CODE GENERATOR\n");
printf("---------------------------\n");
printf("Enter an arithmetic expression per line.\n");
printf("Press Ctrl+D (macOS/Linux) to exit.\n\n");
yyparse();
return 0;
}
Problems Output Debug Console Terminal Ports
intermediate - Assignment-11
kisna@MacBook-Air-2 Assignment-11 % ./intermediate
Final result is stored in: t7
(a+b)*c
t8 = a + b
t9 = t8 * c
Final result is stored in: t9
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intermediate
lex.yy.c
lexer.l
parser.tab.c
parser.tab.h
parser.y
parser.y Assignment-10
lexer.l Assignment-10
parser.y Assignment-11
lexer.l Assignment-11
Assignment-11 > lexer.l
%{
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include "parser.tab.h"
%}
%%
[0-9]+(\.[0-9]+)? {
yylval.str = strdup(yytext);
return NUMBER;
}
[a-zA-Z_][a-zA-Z0-9_]* {
yylval.str = strdup(yytext);
return ID;
}
[+\-*/()] { return yytext[0]; }
[\t\r]+ { /* Ignore spaces and tabs */ }
\n { return '\n'; }
. { return yytext[0]; }
%%
int yywrap(void)
{
return 1;
}
Problems Output Debug Console Terminal Ports
intermediate - Assignment-11
kisna@MacBook-Air-2 Assignment-11 % ./intermediate
Final result is stored in: t7
(a+b)*c
t8 = a + b
t9 = t8 * c
Final result is stored in: t9
Do you want to install the recommended 'C/C++ Extension Pack' extension from Microsoft for the C language?
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lex.yy.c
lexer.l
parser.tab.c
parser.tab.h
parser.y
Assignment-11 > lexer.l
Problems Output Debug Console Terminal Ports
intermediate - Assignment-11
kisna@MacBook-Air-2 Assignment-10 % cd ..
kisna@MacBook-Air-2 Compiler Construction % cd Assignment-11
kisna@MacBook-Air-2 Assignment-11 % clera
zsh: command not found: clera
kisna@MacBook-Air-2 Assignment-11 % clear
kisna@MacBook-Air-2 Assignment-11 % bison -d parser.y
kisna@MacBook-Air-2 Assignment-11 % flex lexer.l
kisna@MacBook-Air-2 Assignment-11 % gcc parser.tab.c lex.yy.c -o intermediate
kisna@MacBook-Air-2 Assignment-11 % ./intermediate
INTERMEDIATE CODE GENERATOR
Enter an arithmetic expression per line.
Press Ctrl+D (macOS/Linux) to exit.
a+b
t1 = a + b
Final result is stored in: t1
a+b*c
t2 = b * c
t3 = a + t2
Final result is stored in: t3
a+b*c-d/e
t4 = b * c
t5 = a + t4
t6 = d / e
t7 = t5 - t6
Final result is stored in: t7
(a+b)*c
t8 = a + b
t9 = t8 * c
Final result is stored in: t9
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